Online Shoppers Purchasing Intention Analysis

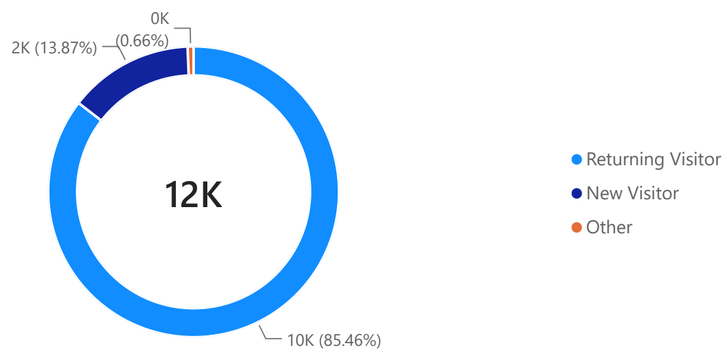
Understanding customer behaviour and factors associated with e-commerce conversion

Total Sessions	Purchased Sessions	Conversion Rate	Average Page Visits
12K	2K	15.6%	34.89

Month

- [ ] Aug
- [ ] Dec
- [ ] Feb
- [ ] Jul
- [ ] June
- [ ] Mar
- [ ] May
- [ ] Nov
- [ ] Oct
- [ ] Sep

Total Sessions
by Visitor_Segment



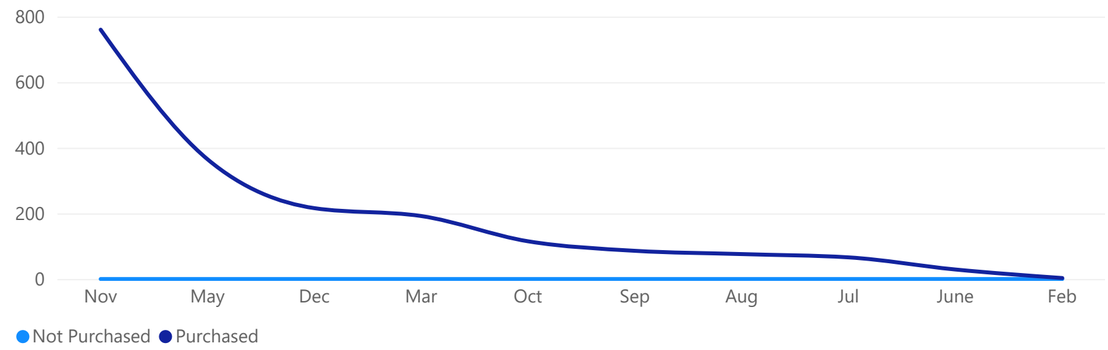
Visitor_Segment

- [ ] New Visitor
- [ ] Other
- [ ] Returning Visitor

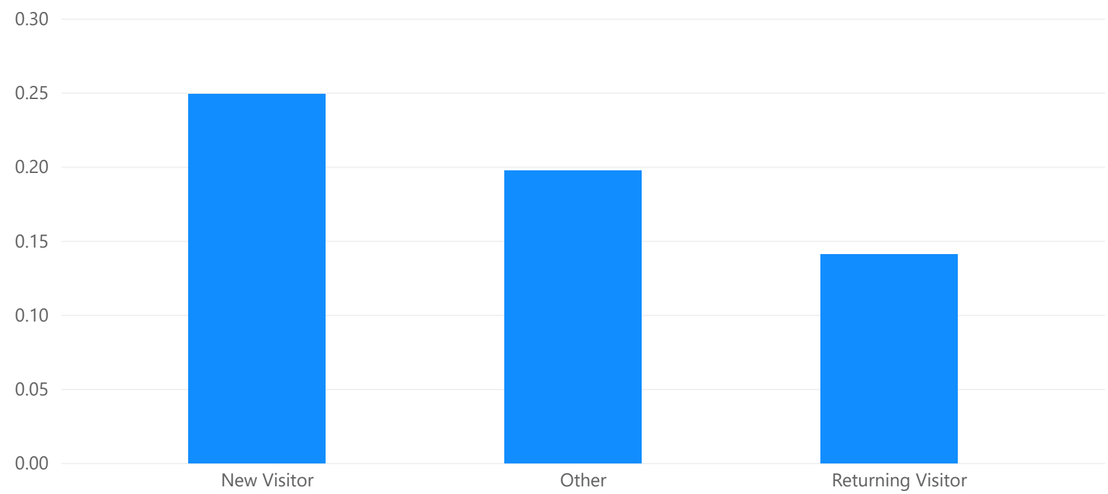
Weekend

- [ ] False
- [ ] True

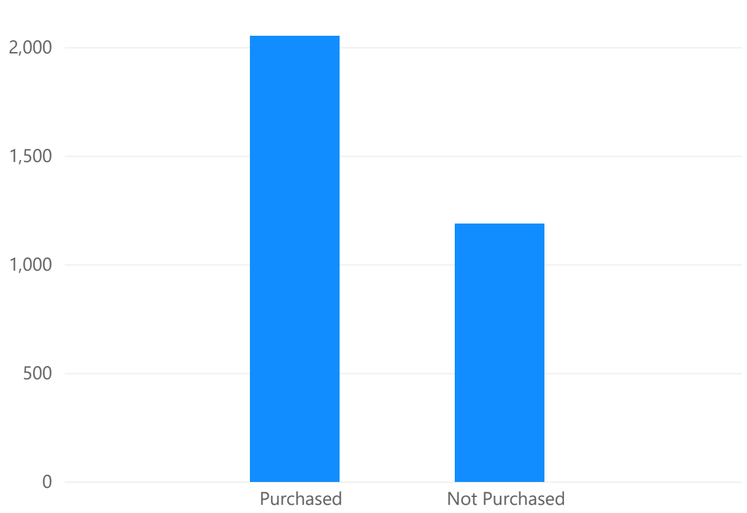
Purchased Sessions
by Month and Purchase_Status



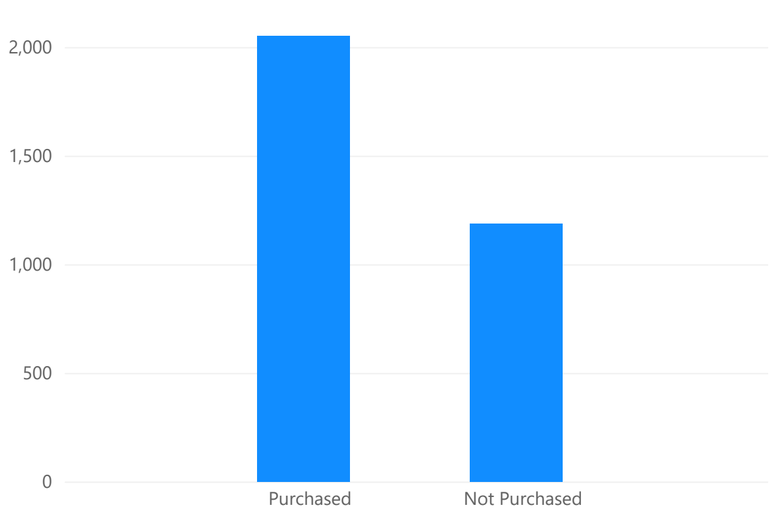
Conversion Rate
by Visitor_Segment



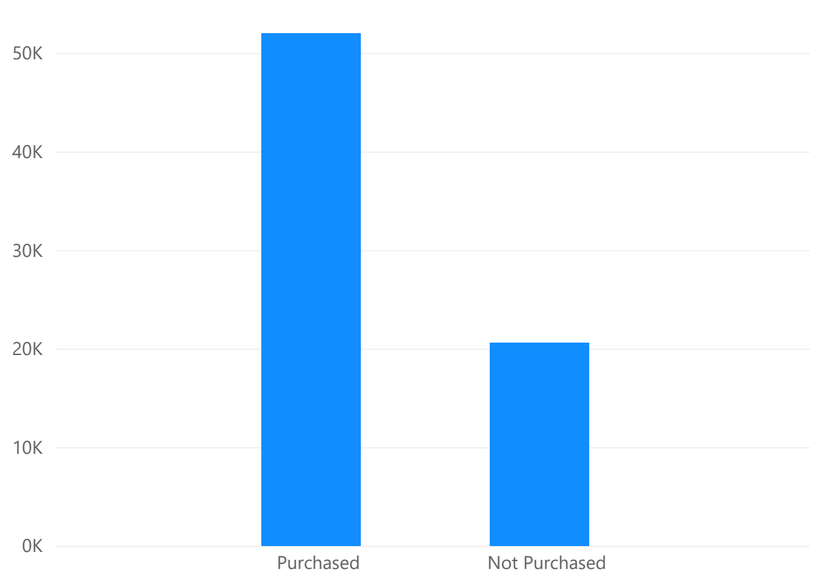
Average Page Visits by Purchase Status



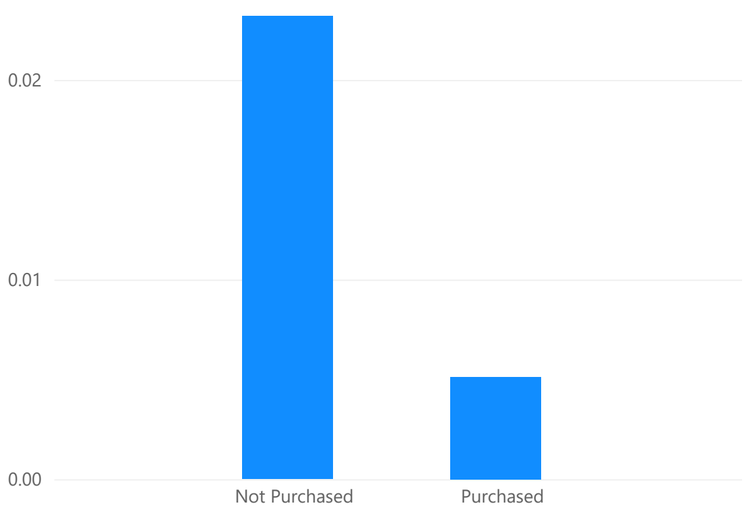
Average Page Duration by Purchase Status



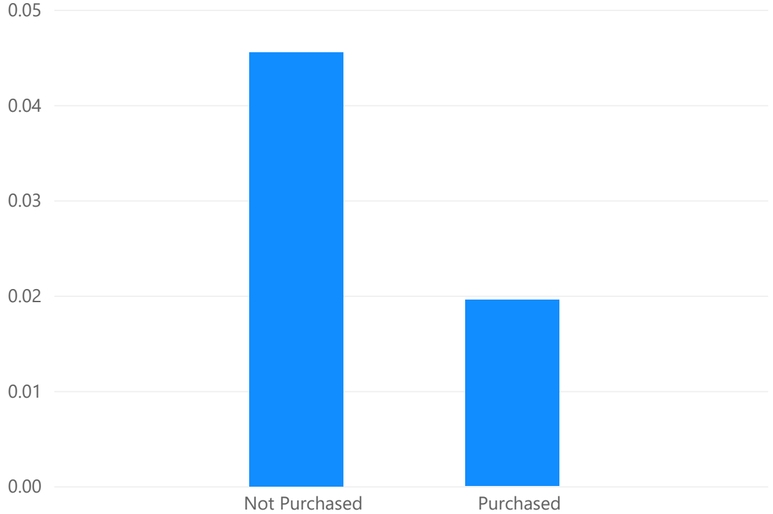
Page Value by Purchase Status



Bounce Rate by Purchase Status



Exit Rate by Purchase Status



Month

- [ ] Aug
- [ ] Dec
- [ ] Feb
- [ ] Jul
- [ ] June
- [ ] Mar
- [ ] May
- [ ] Nov
- [ ] Oct
- [ ] Sep

Visitor_Segment

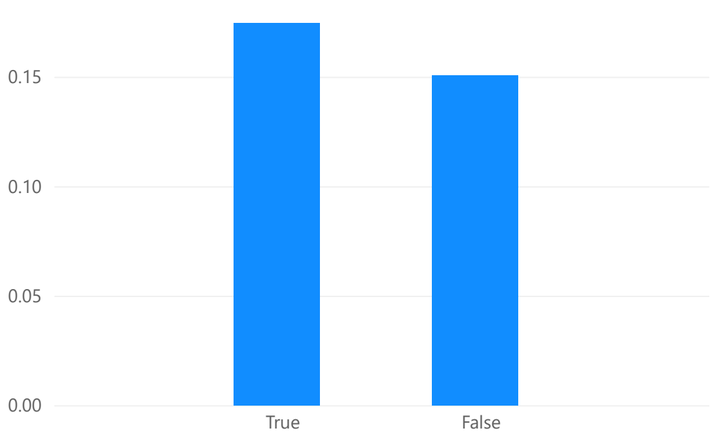
- [ ] New Visitor
- [ ] Other
- [ ] Returning Visitor

Weekend

- [ ] False
- [ ] True

Visitor_Segment	Total Sessions	Purchased Sessions	Conversion Rate	Average Page Visits	Average Page Duration
New Visitor	1693	422	0.25	20.95	747.98
Other	81	16	0.20	14.77	676.63
Returning Visitor	10431	1470	0.14	37.31	1,421.88
Total	**12205**	**1908**	**0.16**	**34.89**	**1,323.45**

Conversion Rate: Weekday vs Weekend



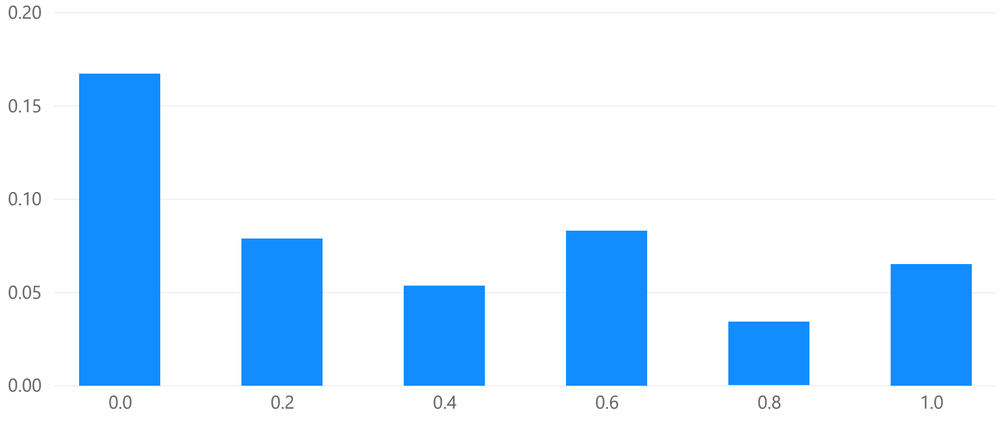
Visitor_Segment

- [ ] New Visitor
- [ ] Other
- [ ] Returning Visitor

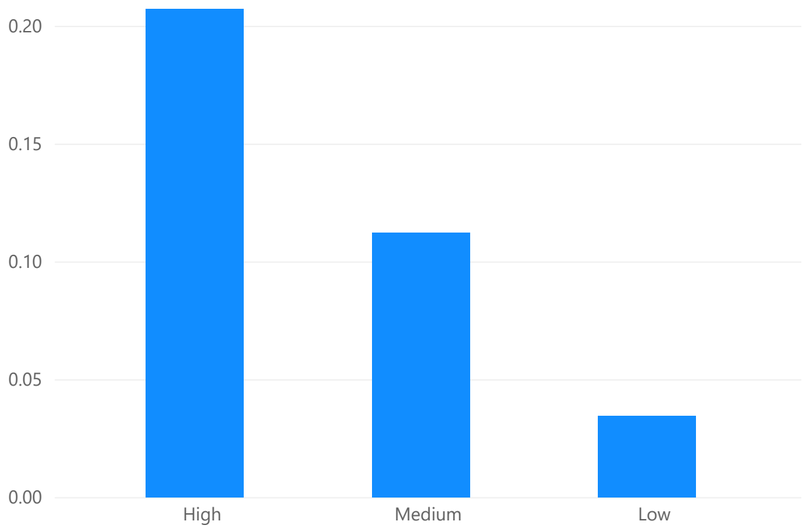
Weekend

- [ ] False
- [ ] True

Conversion Rate by Special-Day Proximity



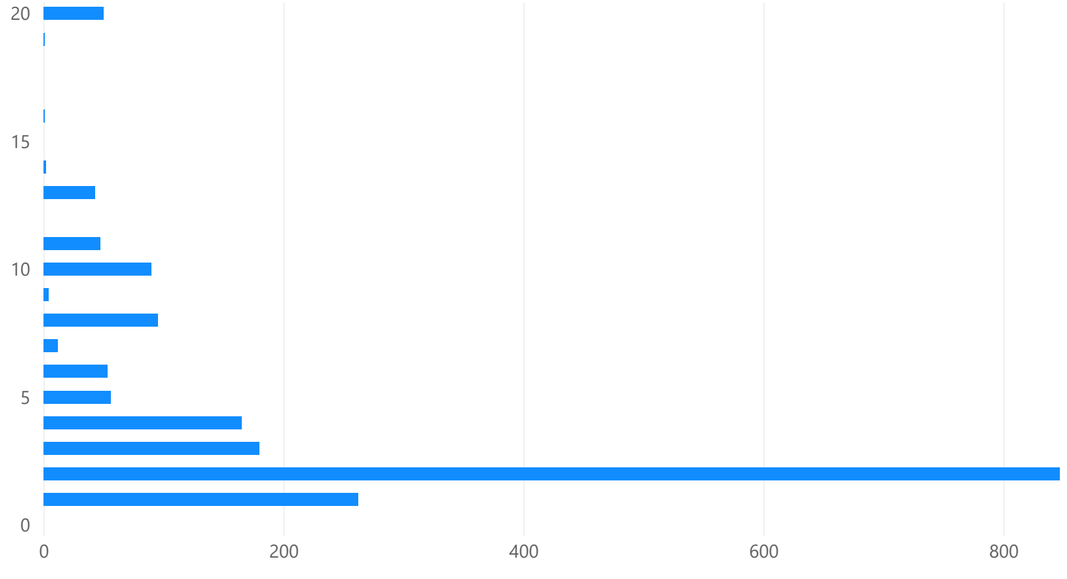
Conversion Rate by Engagement Level



Month

- [ ] Aug
- [ ] Dec
- [ ] Feb
- [ ] Jul
- [ ] June
- [ ] Mar
- [ ] May
- [ ] Nov
- [ ] Oct
- [ ] Sep

Purchases by Traffic Type



INSIGHTS

Insight 1 — Visitor Type

Finding:

New visitors have a higher conversion rate than returning visitors in this dataset.

- New Visitor: **24.9%**
- Other: **18.8%**
- Returning Visitor: **13.9%**

Interpretation

Although returning visitors represent the majority of sessions, their conversion rate is lower than that of new visitors.

Insight 2 — Engagement

Purchased sessions show substantially higher engagement.

Metric	Purchased	Not Purchased
Total page visits	**52.4**	**31.3**
Total page duration	**2,053**	**1,174**

Interpretation

Sessions resulting in purchases were associated with more pages viewed and more time spent across the tracked page categories.

Insight 3 — Bounce & Exit Rates

Purchased sessions had lower average:

- Bounce Rate: **0.51%**
- Exit Rate: **1.96%**

compared with non-purchased sessions:

- Bounce Rate: **2.53%**
- Exit Rate: **4.74%**

Insight 4 — Page Value

Average Page Value was substantially higher for purchased sessions:

- Purchased: **27.26**
- Not Purchased: **1.98**

Insight 5 — Weekend Conversion

Weekend sessions had a higher conversion rate:

- Weekend: **17.4%**
- Weekday: **14.9%**

Monthly insight

From the original data, November has the highest conversion rate among the listed months:

November — 25.4%

Other relatively high months include:

- October — 20.9%
- September — 19.2%
- August — 17.6%

February is particularly low at approximately **1.6%**, although it also has a much smaller number of sessions.

RECOMMENDATIONS

1. Improve returning-customer conversion

Analyze the returning visitor journey and identify potential friction points.

2. Increase meaningful engagement

Improve product discovery, navigation and product information to encourage deeper browsing.

3. Reduce bounce and exit behaviour

Identify pages with higher bounce/exit rates and test improvements to layout, content and calls-to-action.

4. Leverage high-conversion periods

Investigate successful months and evaluate whether similar campaign strategies can be replicated.

5. Optimize traffic sources

Compare traffic sources using both traffic volume and conversion rate, rather than focusing only on the number of visitors.

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